Additional Experiments: Square

In addition to our simulation experiments, we included an additional domain, Square, to assess PHIRL's capabilities.

5.1. Domain and Dataset

Domains We selected an additional domain for Robomimic (Mandlekar et al., 2021): Square. The goal is to control a robotic arm to pick up a square-shaped nut and place it onto a peg, shown in Figure 7.

Figure 7. Robomimic Square Task. The goal is to control a robotic arm to pick up a square-shaped nut and place it onto a peg.

Datasets The dataset setup is the same as in Section 4.

5.2. Baselines and Deployment Details

Baselines The baselines and setups are the same as in Section 4.

Data Collection We were unfortunately only able to recruit two participants for this domain. Thus, the annotation and feedback were from two participants instead of three. Other than that, the setups are the same as in Section 4.

5.3. Results

The results of reward learning experiments are presented in Figure 3.

The top row presents the average environmental return rewards of the last checkpoint across all baselines; the middle row details the success rate for each method; and the bottom row provides the learning curves for all the baselines. The columns, from left to right, correspond to the datasets: PH and MH. Since Pref does not learn from demonstrations, we used the same data for PH and MH conditions. The shaded areas are confidence intervals. The error bars represent the standard deviation, and the reported rewards are smoothed by averaging over 100 evaluation rollouts executed with the best checkpoint using a set of seeds (1 to 100).

Across both datasets, despite the success rates are relatively low, PHIRL and PHIRL-online are the only methods that achieved task success. With the PH dataset, PHIRL-online significantly improved the task performance compared to PHIRL (39 success vs. 24 success), while there was no improvement with the MH dataset. This is contradictory with our previous observations (PHIRL-online would improve the performance more if the demonstration quality is low). We hypothesize that this is because most failures happened when the robot dropped the square onto the peg, and the overlap of the square and the peg creates out-of-distribution visual data, which leads the VLM to produce unreliable progress labels (PHIRL-online's performance in both datasets shows a larger variance). We will validate this assumption by re-creating specific scenarios and checking the values from the VLM. If the statement is true, we will update the paper and include it in the discussion.

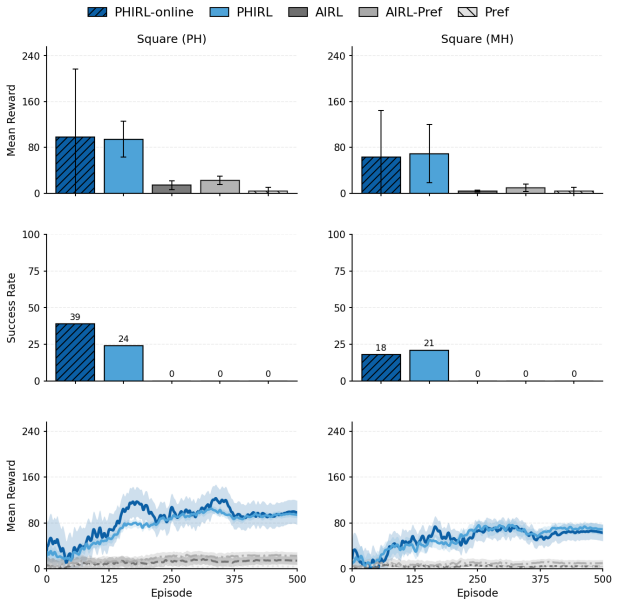

Figure 8. Square Results. First row: average return rewards; second row: success rate over 100 runs; third row: reward curves. The error bars in row one are standard deviations. The shaded areas in row three are confidence intervals. PHIRL and PHIRL-online are the only methods that achieved task success.